AgriConnect

FARMER MARKETPLACE SYSTEM

React . Spring Boot . MySQL . JWT

SYSTEM DESIGN ASSIGNMENT

UML Diagrams for AgriConnect

Class | Component | Deployment | Use Case

Name	**Sonam Rai**
Roll Number	**2501560005**
Course	**MCA**
Specialization	**MCA Core**
Semester	**3rd Semester**
Problem Statement	**AgriConnect - Farmer Marketplace System**

Submitted for Internal Evaluation - System Design

1 Problem Statement

AgriConnect is a farmer marketplace system designed to connect farmers, field coordinators, buyers and administrators through a digital platform. Farmers can create crop listings with quantity, price, location and images. Buyers can search and filter available crops and place orders. Field Coordinators can collect farmer information from the field and add crop listings on behalf of collected farmers, even when the farmer does not have a registered account. Administrators manage users and marketplace data.

Actors in the system

Farmer

Creates and manages crop listings with quantity, price, location and images.

Buyer

Searches and filters available crops and places orders.

Field Coordinator

Collects farmer data in the field and lists crops for unregistered farmers.

Admin

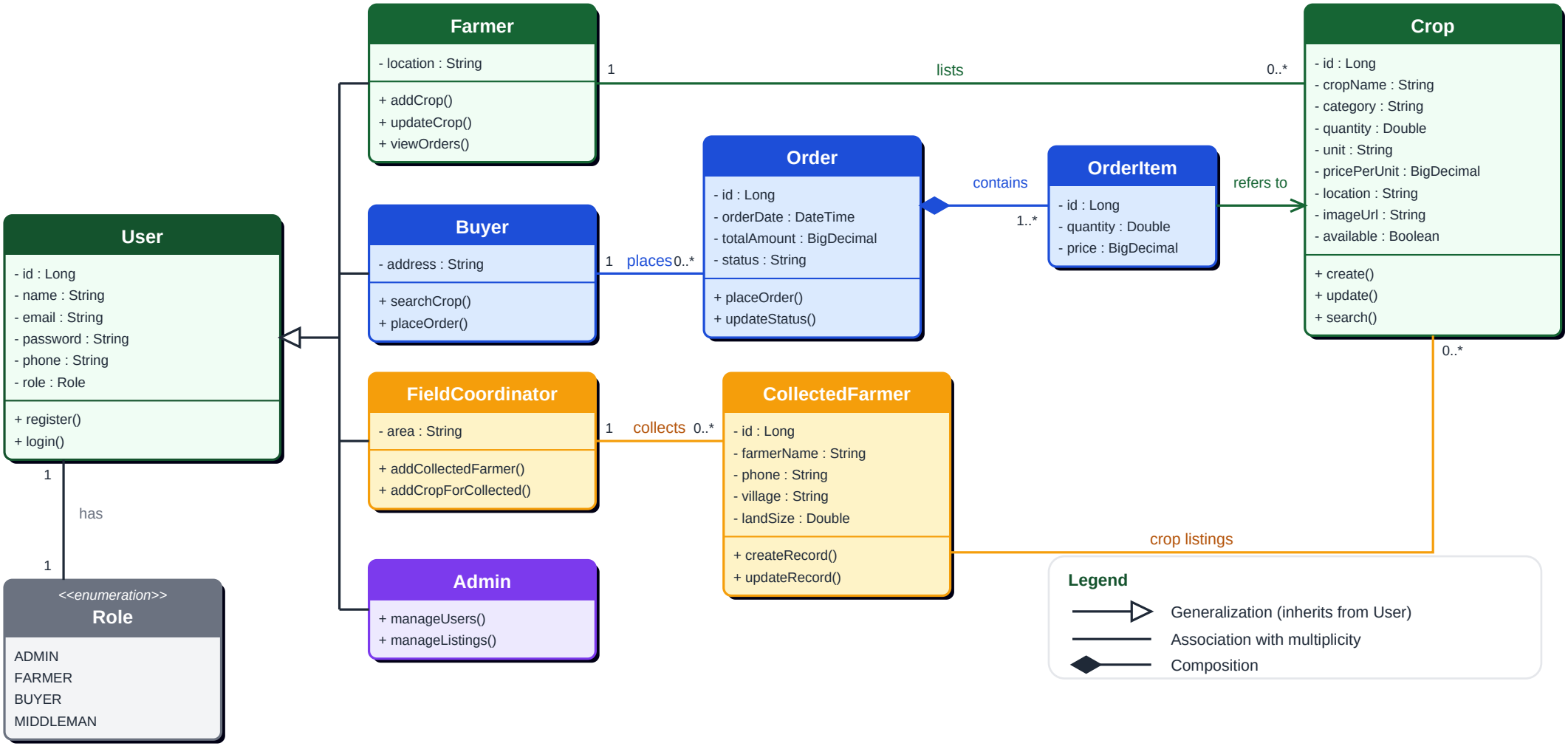
Manages users and marketplace data across the platform.

Technology stack

React	**Spring Boot**	**Spring Data JPA**	**MySQL**	**JWT**	**Role-based access**
Frontend	REST backend	Persistence	Database	Authentication	Authorization

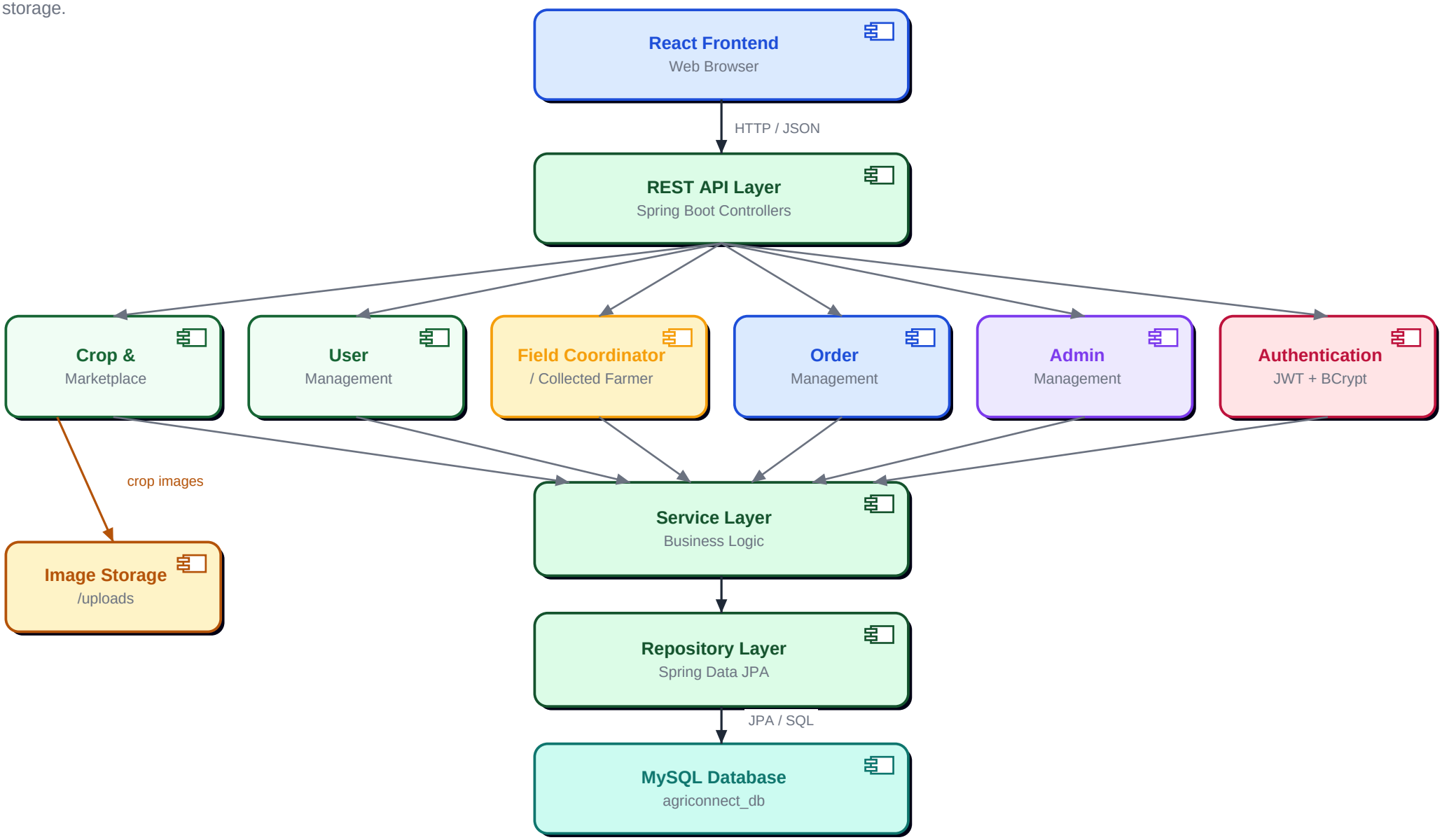
2 Class Diagram

The class diagram represents the main static structure of AgriConnect, including users, roles, crop listings, collected farmer records and orders.



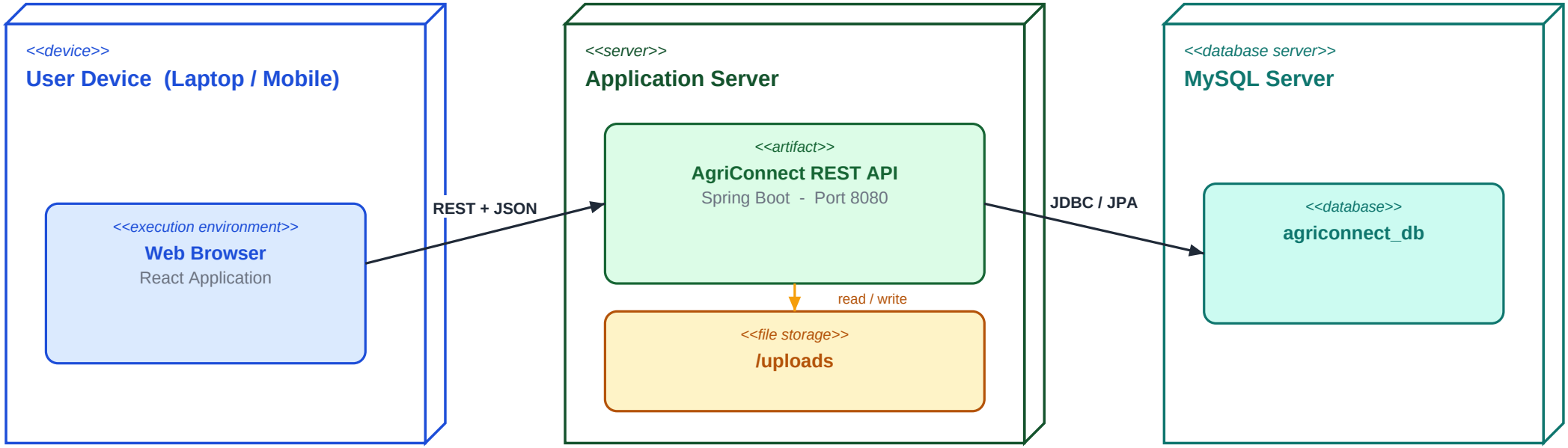
3 Component Diagram

The component diagram shows the major software modules and their dependencies: React frontend, REST API/controllers, authentication, services, repositories, MySQL and image storage.



4 Deployment Diagram

The deployment diagram shows the physical/runtime environment: user device and browser, Spring Boot application server, MySQL database server and file storage.

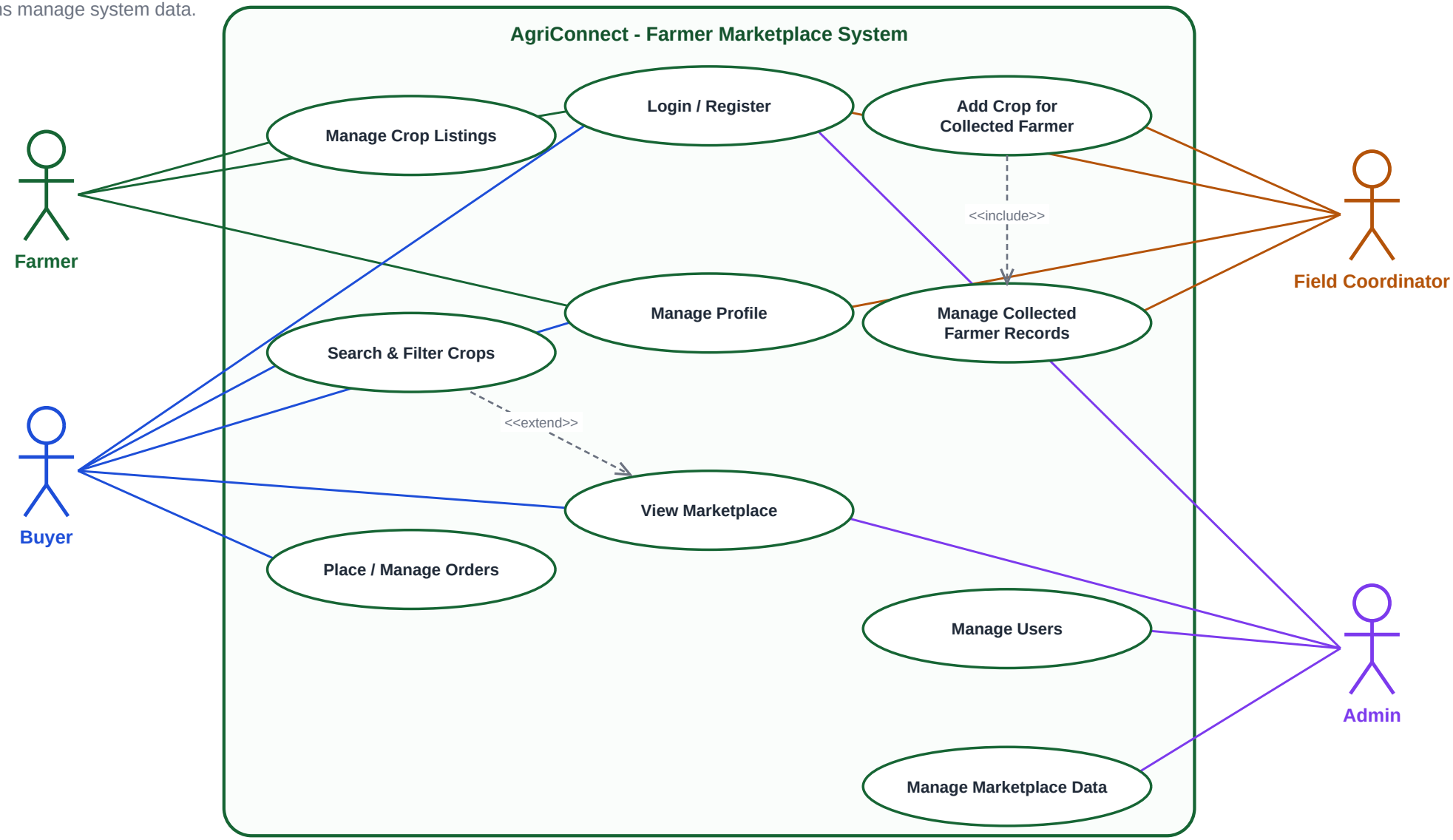


Runtime flow

1. The user opens the React application in a web browser on a laptop or mobile device.
2. The browser calls the Spring Boot REST API (port 8080) using JSON over HTTP.
3. The API reads and writes crop images in the /uploads file storage.
4. The API persists all data in the agriconnect_db database through JDBC / JPA.

5 Use Case Diagram

The use case diagram identifies the actors and their main activities in AgriConnect. Farmers manage crops, Buyers search and order, Field Coordinators maintain field records and Admins manage system data.



6 Conclusion

These UML diagrams describe the **structure**, **components**, **deployment environment** and **functional interactions** of the AgriConnect system. They provide the required System Design documentation for internal evaluation.

Class Diagram

Static structure: users, roles, crops, orders and collected farmers

Component Diagram

Layered modules from React UI to MySQL and image storage

Deployment Diagram

Runtime nodes: device, application server, database server

Use Case Diagram

Actors and their main activities in the system

Student: Sonam Rai

Course: MCA (MCA Core) | Semester: 3rd Semester | Roll Number: 2501560005